

Reflective Journal – Lab 7: Controlled Generation Study

Overview

Lab 7 was about experimenting with a text-to-image pipeline and learning how different parameters affect the results. Instead of chasing perfect pictures, the focus was on judgment — seeing how guidance scale, sampling steps, seeds, and negative prompts change outputs, and recognizing when the model fails. Editing one of my own photos and deliberately triggering a failure mode made the lab feel hands-on and personal.

Key Learnings

- **Guidance Scale:** Higher values forced the model closer to the prompt, while lower values allowed more creative variation.
- **Sampling Steps:** More steps improved detail, but after a point the gains weren't worth the extra time.
- **Seeds:** Locking a seed gave reproducible outputs, while changing seeds showed how randomness drives variation.
- **Negative Prompts:** Adding a negative prompt was a simple way to suppress unwanted features.
- **Failure Modes:** Triggering a known failure (like distorted hands) helped me explain why these models break down.

Application to Practice

What stood out to me is how these same controls apply in real scenarios outside the lab. For example, reproducibility through seeds matters if someone is generating synthetic training data for testing a vision model. Guidance scale and negative prompts are useful when creating clear visuals for communication — like marketing materials, design mockups, or even educational diagrams. Understanding failure modes is important in any industry that uses generative AI, because it prevents people from blindly trusting outputs that might be misleading. In short, the lab showed me that controlled generation isn't just about art; it's about knowing how to steer a model responsibly in whatever context it's being used.

Reflection

This lab taught me that generative AI is less about “magic” and more about controlled experimentation. By changing one setting at a time, I learned how each parameter influences the final image. Editing my own photo made the process feel personal, and deliberately causing a failure reminded me that these systems are far from perfect. The biggest takeaway is that judgment matters: knowing when to trust an output, when to rerun with a fixed seed, and when to recognize that the model simply can't handle a certain prompt. Lab 7 gave me confidence in using generative tools critically, not just creatively.